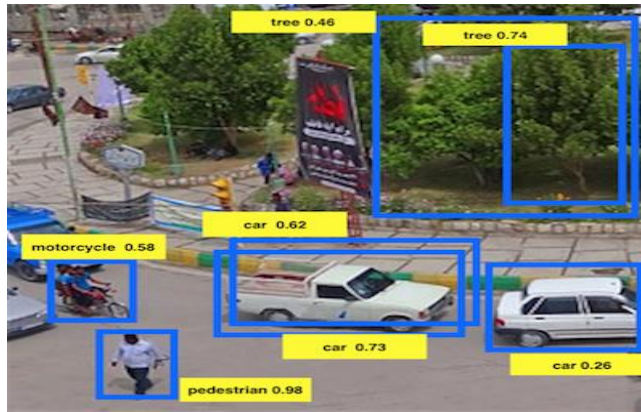




Question 1: [44 Marks]

A) Choose the correct answer (use tabular format):

1. In order to match two images with different scale, which feature detector should be used?
 - a) Harris
 - b) SIFT
 - c) both works well
 - d) none of them
2. Computing the transformation between two translated images using RANSAC requires at least pair(s) of matching interest points.
 - a) one
 - b) two
 - c) three
 - d) four
3. In Harris detector, the decomposition of the eigenvalues λ_1 and λ_2 of the second moment matrix M reveals the existence of a corner when:
 - a) λ_1 and λ_2 are small
 - b) $\lambda_1 \gg \lambda_2$
 - c) λ_1 and λ_2 are large
 - d) $\lambda_2 \gg \lambda_1$
4. You are building a 3-class object classification and localization algorithm. The classes are: pedestrian (c=1), car (c=2), motorcycle (c=3). What would be the label for the following image? Recall $y=[p_c, b_x, b_y, b_h, b_w, c_1, c_2, c_3]$
 - a) $y = [0, 0.3, 0.7, 0.3, 0.3, 0, 1, 0]$
 - b) $y = [1, 0.3, 0.7, 0.3, 0.3, 0, 1, 0]$
 - c) $y = [1, 0.3, 0.7, 0.5, 0.5, 1, 0, 0]$
 - d) $y = [0, 0.2, 0.4, 0.5, 0.5, 0, 1, 0]$
5. Suppose two bounding boxes with sizes 2x2 and 2x3 are overlapping in a region of size 1x1. The IoU between these two boxes will equal
 - a) 1/6
 - b) 1/9
 - c) 1/10
 - d) none of them
6. Consider the Siamese network architecture for face verification task, the upper and lower neural networks should have input images, and parameters.
 - a) different / different
 - b) different / don't care
 - c) same / different
 - d) different / same
7. The homography/transformation matrix H describes the transformation between points in two related images from different
 - a) rotation
 - b) aspect
 - c) shear
 - d) perspective
8. In order to discriminate a person in an image, which feature detector works better?
 - a) Harris
 - b) HOG
 - c) SIFT
 - d) none of them
9. The up-sampling operation in a convolutional neural network can be implemented using
 - a) pooling
 - b) strided convolution
 - c) transpose convolution
 - d) deconvolution
10. Suppose you run non-max suppression on the predicted boxes below. The parameters you use for non-max suppression are that boxes with probability ≤ 0.4 are discarded, and the IoU threshold for deciding if two boxes overlap is 0.5. How many boxes will remain after non-max suppression?



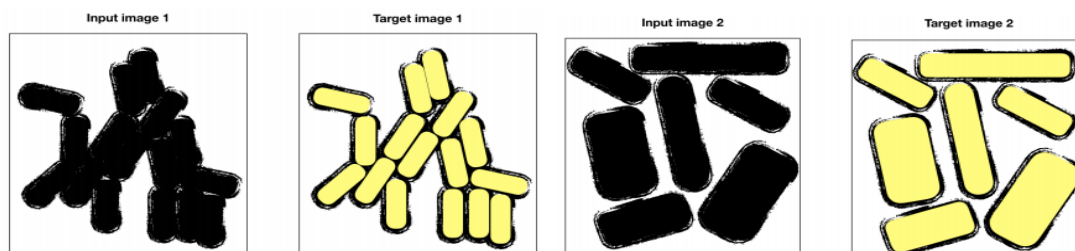
- a) 3 b) 4 c) 5 d) 6
11. Suppose you are using YOLO on a 19x19 grid, on a detection problem with 20 classes, and with 5 anchor boxes. During training, for each image you will need to construct an output volume y as the target value for the neural network; this corresponds to the last layer of the neural network. The dimension of this output volume will be
 - a) 19x19x45 b) 19x19x125
 - c) 19x19x500 d) none of them
 12. The intuition behind detecting a corner in an image is that if the corner window slides for a small shift, the intensity change of the window pixel values caused by the shift is
 - a) low b) moderate c) high d) doesn't change
 13. In a Siamese network, the definition of $\max(\|f(A) - f(P)\|_2^2 - \|f(A) - f(N)\|_2^2 + \alpha, 0)$ would be a correct definition for a triplet loss (where A represents the anchor image, P represents the positive image and N represents the negative image), considering that
 - a) $\alpha = 0$ b) $\alpha < 0$ c) $\alpha > 0$ d) $\alpha \geq 0$
 14. Consider using k-means for color image clustering, if you want pixels of similar color but in discontinuous regions to be clustered in different clusters, what kind of feature(s) you may use as input to the k-means algorithm?
 - a) Pixel color b) Pixel location
 - c) Both color and location d) Inapplicable using k-means
 15. All of these metrics can be used for accuracy evaluation of multiple object tracking except
 - a) Mostly Tracked (MT)
 - b) Frame per second (fps)
 - c) Average IoU
 - d) Multiple Object Tracking Precision (MOTP)
 16. The final convergence of the MeanShift clustering algorithm depends on
 - a) window initial position b) outliers existence
 - c) features dimension d) window size
 17. In an object detection model where it is supposed to produce a bounding box for each object that appears in the image, you consider a detection output as a false negative if
 - a) $\text{IoU} > 0.5$ b) $\text{IoU} < 0.5$
 - c) $\text{IoU} = 0.5$ d) no bounding box produced for the object
 18. Using RANSAC for fitting a translational model, suppose we know that 30% of our data is outliers. How many times do we need to sample to assure with probability 80% that we have at least one sample being all inliers?
 - a) 1 b) 1.3
 - c) 2 d) 3

19. Which problem the Hierarchical LK optical flow algorithm tries to solve?
 - a) The aperture problem
 - b) When the brightness constancy constraint is no longer satisfied
 - c) When the motion is no longer small
 - d) When the nearby pixels do not move together
20. In the Bag of Words (BOW) algorithm used for local feature indexing in the context of object recognition from images, the number of k clusters selected by the k-means algorithm should be:
 - a) representative to the number of visual words in the image dataset
 - b) representative to the number of images in the dataset
 - c) representative to the number of classes in the image dataset
 - d) representative to the number of features extracted from the image dataset
21. A good feature descriptor should be
 - a) repeatable
 - b) distinctive
 - c) local
 - d) all of the above
22. The task of determining the class label assigned to video sequence is defined as
 - a) object detection
 - b) action recognition
 - c) object tracking
 - d) video localization

Question 2:

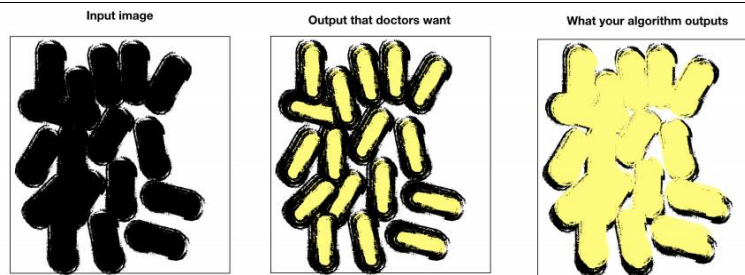
[13 Marks]

- A) [7 Marks] Assume that you have a video of the street filmed from a static camera where there is a car you need to track. Assume that you have a color model of the car that you are tracking. The model is described as a color histogram that has 16 equally-spaced bins of (only) the hue channel in HSV color space. Explain how this color model can be used to track the car along the sequence of frames in the video. (Hint: use MeanShift algorithm)
- B) [6 Marks] You have been hired by a group of health-care researchers to solve one of their major challenges dealing with cell images: determining which parts of a microscope image corresponds to which individual cells. Given examples of input images and the corresponding target images as follows, the input images are taken from a microscope, and the target images have been created by the doctors. Your task is to implement an automatic model to achieve this task, where each pixel of the target image is labeled as 0 (this pixel is not part of a cell) or 1 (this pixel is part of a cell).



- i. Name the title of this computer vision task.
- ii. Design a network based on CNN to implement it (Sketch a general architecture diagram in terms of input, core and output components).

After finishing your model, the doctors are very unhappy, and argue that they would like to visually distinguish cells. They show you the following prediction output, which does not clearly separate distinct cells. Describe how you would correct your model to satisfy the doctors request.



- iii. Mention the title of this modified computer vision task.
- iv. Suggest the needed modification in the target images provided by the doctors in order to support the training of your model.

Question 3:

[10 Marks]

We want to build a safety system for cars that detects pedestrians from a single camera image. We have a large dataset of images with pedestrians labelled by a bounding box. We want to be able to detect and locate pedestrians regardless of their distance and apparent size in the image.

- i. Compare between R-CNN, Fast R-CNN, Faster R-CNN and YOLO object detection and localization algorithms in terms of (use tabular format):
 1. Used method of region proposals.
 2. Number of ConvNets used for feature extraction of input image.
 3. Object classification function.
 4. Speed.
- ii. Given a video sequence of a city traffic with moving pedestrians, a Multiple Object Tracking (MOT) system can be designed to track all pedestrians in the video. The tracker can get benefit from a pedestrians' detector, where the detector is responsible of detecting pedestrians in each frame, and the tracker is responsible of associating detections from time frame t to frame $t+1$.

Define what kind of error(s) this tracker should handle for each of the following scenarios in the video sequence:

1. Many partial/complete occlusions of pedestrians occur.
2. Many pedestrians cross each other.

Question 4:

[8 Marks]

In the domain of face recognition, suppose that the face images all occur roughly on a line in the high dimensional space of images. If the image is represented in grayscale intensity, an 100×100 image is a point in 10,000 dimension. Describe a model for face recognition based on PCA (dimensionality reduction) that works by compressing the images to a low-dimensional subspace which captures the key appearance characteristics of the visual faces.

- i. State the steps of both Training and Testing procedures.
- ii. What are the strengths and the weaknesses of this type of representation for human faces?

😊 Wish You Good Health & Happy Graduation 😊

Dr. Dina Khattab